ActInf ArtStream 003.1 ~ Diana Omigie: "A model of time-varying music en

ArtStream | Diana Omigie | 1:15:27

Hello and welcome. It's September 9th, 2024. This is Active Inference Art Stream number 3.1 with Diana Omegi talking about curious sounds, prediction, and the human drive to engage with music. There will be a presentation and then we'll read some comments and questions from the live chat. So thank you, Diana, for joining and looking forward to your presentation.
Thank you, Daniel, for the invitation and to everyone who's joined today. I'm a senior lecturer and researcher at Goldsmiths University of London. And today I'll be talking, giving a talk that's entitled Curious Sounds, Prediction, and the Human Drive to Engage with Music. And the reason I chose this title was because it recently came across this statistic. The International Federation of the Phonographic Industry do a report every year where they say how much the average person listens to music. And it turns out it's about 20.7 hours a week. So if we assume that people get about eight hours of sleep a night, then we're looking at about a fifth of our waking hours that the average person is listening to music. And it struck me that there's a really interesting question here, which is what is behind this drive to engage with music at this scale? And so what I'd like to do in today's talk, partly because I wasn't quite sure who would be attending, but I assumed it would be people with lots of different backgrounds. And yeah, what I decided to do was just present a little bit about what neuroscientists have been finding out about the neural substrates of music perception and pleasure. We might assume that has something to do with why we choose to listen to music. I then want to talk for a few minutes about what the predictive processing framework could explain to us or tell us about why it is that our attention towards stimuli might change over time. And finally, use what we've learned from neuroscience, from the predictive processing coding framework to present a model, basically, of the dynamics of extended engagement with music. So that's a piece of work that was published earlier this year and which I'm keen to share with interested parties. Let's think about music perception and the things that we do without even realizing how powerful these things are. So we can spot wrong notes in music that we're hearing for the first time. By wrong, I mean, yeah, notes that were clearly not intended by the performer. We can say whether two melodies are the same or different, again, hold pieces of sequences of pitches and rhythms in our memory and compare them.
We can recognize familiar pieces, even if they've been played differently, different lyrics, different instruments and so on and so forth. And we can correctly predict how a piece of music we've never heard before might unfold. We might be able to sing along to it, even if we're hearing it for the first time. And naturally, people have come up with ideas as to what's going on, why this is all possible, how this is all possible. For example, in this model published in 2003, the idea was this acoustic input comes in undergoes some analysis, inevitably gets separated out into its sort of pitch components and temporal components that undergo further processing these interface with some store of musical phrases and patterns that we've experienced in the past and so on and so forth.
But of course, this is a sort of box and arrow model of what makes sense as a series of

processes. When we look at the brain, we're looking at a network of connections between subcortical structures, the auditory cortex, so-called belt and power belt areas around the auditory cortex.
And we're looking beyond that, we're looking at sort of more frontal parts of the brain that we know are involved in some of the more complex functions that we're able to do as humans. And what's interesting is that when it comes to something like pitch perception, which is obviously a very important part of processing music in lots and lots of different cultures, we see that the electrophysiological responses to pitch, in this case measured intracranially, are very much in line with the principles of predictive coding. In other words, it's been shown that early tonotopic structures feed information about frequency to later areas where we seem to have this abstract model of pitch being formed and predictive signals being sent back. So we're seeing something as fundamental as just determining what pitch we're hearing. Not something as fundamental, not surprising, something as fundamental as pitch perception seeming to have this, follow this predictive coding principle.
Most of us can do lots of things with music, recognize it, discriminate pieces of music. Not everybody is able to do these everyday tasks. So for over 200 years, or almost 200 years, there have been reports or just papers about what we now call congenital amusia, difficulty making sense of music that can manifest as difficulty discriminating melodies and recognizing melodies that should be familiar, all those things I said we do so naturally. And it's not surprising that when this was begun to be researched in the early 2000s that it was considered perhaps a disorder of fine-grained pitch discrimination. Some of the tasks suggested that the individuals who had it were struggling to say whether a pitch had, you know, changed if it was not a very large interval that was being presented. However, as more work came out suggesting that this could, that people with generally musia may also have difficulties for pitch memory or with pitch memory, the question that arose is whether there might be a more general mechanism that's sort of compromised or in this condition. And so that was one of the questions I was asking during my PhD in the late 2000s, where I was lucky to be at the same institution as Marcus Pierce, who'd developed this statistical learning model of musical expectations. So it's a statistical learning model, which means that it learns from the regularities and the corpus of music you give it, and a piece of music as it unfolds, and is able to spit out these values for any given event in that piece of music. So information content, how surprising an event, a note should be to a listener, about what that incoming note will be. And using this model was able to sort of quantify the surprise in every single note that we were presenting, well, in the melodies that we were presenting to our participants with amusia and participants without, and were able to show that amusics were showing less evidence for sort of prediction error signaling in response to the notes in these melodies. Further work later on from different labs corroborated this using dynamic causal modeling on MEG data showing that there was indeed this abnormal prediction error from auditory to frontal areas in those with amusia. It was shown that if you looked at resting state activity, the degree of connectivity between frontal and temporal areas was much reduced in those with amusia.

And again, around this time it was coming out that if you look to the structural connectivity, the arcuate, the bundle of axons, the binds the temporal and parietal cortex through and through frontal areas, then there was also this, there was a compromise in the structure or integrity of of this bundle of axons suggesting that there was some sort of under-connectivity between the frontal and temporal areas, superior temporal and inferior frontal areas of the brain and amusia, and that this might be materializing or manifesting in abnormal prediction error signaling.
This is work, I mean this idea of STG and IFG needing to communicate for predictions and prediction error signaling was something that had was being more and more supported than just typical listeners in the range of electrophysiology and also fMRI studies. But since you know, you know, those 20 years ago there's been yet more studies showing that it's not just those areas that seem to be tracking prediction error or showing prediction error like responses. We see that areas like the insular cigulate gyrus and insular are also tracking the information content of surprise of musical events, which, you know, is a nice explanation for why these kinds of events in surprising events can lead to sort of physiological arousal and more visceral engagement. And yeah, in other work that we've done, we've shown that again, the size of the prediction error response is reflected, in this case reflected in pupil dilation response, is even modulated by the entropy of the musical context. So it mattered whether or not what someone was listening to was very predictable or not, whether they would show a prediction error response. And those of you who are into active inference will know that this sounds a lot like sort of precision waiting happening. Just a couple more words about music perception. When we're listening, we come with lots of different backgrounds. Some of us or some people have, you know, musical training and therefore sort of musical expertise. They may or may not even have know how to play the piece that they're listening to. And in this study, for example, it was shown that when a musician was listening to a piece that they'd just learned, compared to when they hadn't learned it before, or compared to when someone who didn't know how to play the piece was listening, you saw this frontal parietal motor related network being activated, just speaking to the fact that musical perception isn't just the processing of the acoustic information, but seems to implicate or recruit areas involved in movement and motor planning. And just to point out that yes, it's always the case of, it was very often the case, the musicians will show more of this activity. But most listeners when they are listening to a piece of music will show engagement of these sensory motor areas. And that was just to prepare you for this idea that that's coming out or has been around for a while even. But music perception involves two pathways, much like speech and language perception involves two pathways, a so called ventral stream or pathway, where core auditory areas sort of move their information towards more inferior frontal areas, allowing the formation and storage of sound patterns that can then allow the registration of expectation violations, and a more dorsal stream involving the parietal and premotor, dorsal lateral prefrontal cortices, allowing the temporal and motor manipulation of information that's been stored in working memory and allowing predictions, but not about what one is going to hear, but rather about when one is going to hear to hear it. So it was a bit more temporal predictions. And now just switching to, you know, to say a few words about musical pleasure and what the sort of body of neuroscientific literature

is saying about this, what it's, you know, what people have done in this area, I'd say most of the work has come from the lab of a registrator in McGill University in Canada, where in a very important seminal study, a PhD student in Valerie Salonpour, they showed that, yeah, when we experience chills, these very pleasurable physiological arousing experiences of pleasure during music listening, when we experience them, we see activation of the so-called ventral striatum of the diputminergic network, whereby the sort of anticipation of this chills, sorry, whereas the anticipation of the chills experience recruits this other part of the striatum, which is the dorsal striatum. So in effect showing that the experience of chills the chills is, or pleasure during music, is recruiting and rewarding areas, but in actually quite nuanced ways, and that separates out the anticipation from the actual experience. So those were, that was a study where they got people to say what moments and what pieces of music they experienced these very pleasurable and physiological, the arousing moments. But in, yes, another study where they presented people with music that they'd never heard before, they were still able to show that the pleasure people reported in response to that music was explainable by how much connectivity they were seeing between the superior temporal gyrus and the ventral striatum. And so they really stressed the importance of this connectivity in when we're, you know, talking about musical, musical pleasure. One way to, to see whether this holds true is, of course, to find people who really don't experience pleasure in response to music. And hedonia means, you know, lack of pleasure and not experiencing pleasure. And there's some people who, and it's associated with depression and other sort of conditions. But around the early 2010s, they, these colleagues showed that there might be some people out there who have normal responses, reward responses, or find lots of things in life rewarding, but just not music.
And so these people with so-called music specific and hedonia seem to be able to recognize and understand music, discriminate melodies, you know, predict what's going to happen next, but they failed to enjoy it. And what they were able to show was that indeed, these individuals would show normal responses to secondary rewards, physiological arousal to secondary rewards like money. But indeed, when they were exposed to moments in music, but should, you know, for most people, the experience is, you know, very chilly and very pleasurable and quite arousing. And they just, they were able to show that there was, and hedonia just didn't show those kinds of responses. And importantly, they were able to show that when they looked at connectivity between the STG and ventral striatum, that indeed, this connectivity was much reduced for those with and hedonia compared to those with, you know, regular listening levels of pleasure and those who really love music.
Just to say that in one case study published a year after this functional connectivity study, they showed that someone with severe case of musical anhedonia showed much reduced connectivity between the superior temporal gyrosis and the NACC corroborating this idea that the connectivity between these areas is important for reward. And last but not least on the topic of pleasure, you know, and yes, another way of testing whether this, you know, this network or, you know, that I've been describing is, is important for music listening is to show that within a given individual, you can kind of turn off and on the pleasure they experience from music. And again, work from the very productive tutorial lab showed that if you stimulated the DLPFC, which is believed to stimulate

the striatum, then you'll show then, yeah, the excitation of this, the excitatory sort of protocol would lead to greater liking, willingness to buy a piece of music that's just been heard, and increase in the arousal demonstrated, where conversely, and inhibiting that pathway, so reducing striatal activity seems to reduce pleasure and willingness to buy that piece of music you just heard, and the physiological responses. They were also able to show that when you put people in the scanner immediately after sort of, or around the time they'd just had this, this trans-carineal magnetic stimulation applied, yes, it was the way to show that the degree of connectivity between right auditory cortex and ventral striatum, again, just this auditory cortex reward area connectivity, predicted the extent to which they had shown enhancement of pleasure from, from TMS excitation. In other words, showing this link between, you know, confirming that the DLPFC was probably stimulating dorsals, sorry, striatum, which was leading to more or less connectivity with the STG, which was leading to more or less pleasure.

But just around this section of my talk up, what I've tried to do is show that when we're listening to music, the acoustic input is going through a series of areas that are carrying out processing at very different, at different stages, and especially when we get to sort of the inferior frontal gyrus and superior temporal gyrus, we're looking at, we're seeing the neural substrates of prediction, predictions and prediction error signaling, and this seems to be compromised in those within musia, and that might explain the wide-ranging difficulties with music. When we come to speaking about pleasure, inevitably, people speak about the role of the auditory system and its, you know, capacity for prediction, but they point out that most important is this, more important, say, even than the STG and frontal temporal, or frontal area connectivity is the, you know, temporal areas involved in auditory processing and the reward areas, and this seems to be compromised in those who report much reduced pleasure from music. And I can really recommend a book called, it was called, but it was published by Zatorre in 2024, where he does a really lovely overview of the work that's come, lots of work that's come out of his lab and lab of, in several labs around the world, trying to understand, you know, put together what we know about music perception, how that sort of interfaces with pleasure, and beginning to tell us a story around the role of prediction, prediction, prediction, yeah. But just to say that, you know, my interest has always been trying to think about music listening as more than just emotions or pleasure. These are obviously things that are very important to us, but I've always had this desire to understand the temporal dynamics of our engagement with it, and, you know, studies have shown, and these have only come out since I've been pondering this, that as with many activities, you know, we will mind wander a large proportion of the time, you know, the time that we think we're doing that activity, in this case music, listening, and, you know, for most tracks, our engagement with a piece of music will rain or just, you know, drop by quite a bit within a few seconds. And, it's clear that music can attract our attention, it can keep us listening, but it can also lose us, sometimes it will catch, attract our attention again, and the question is when and why are we listening, and when and why are we not listening?

And I thought this is where predictive coding framework and the sort of active inference could maybe help us, you know, further our thinking on this topic.

And, and so just to sort of talk about music listening and the dynamics of well, talk about the dynamics of potential engagement in these terms.

If we think about the auditory system and engagement in the auditory realm, then, well, in general, predictions in the predicting coding framework are set to ascend this hierarchy that make for better models that in turn allow us to make better predictions.

And the auditory system, this is, you know, prediction errors ascending from the thalamus to the auditory cortex, where posterior expectations, you know, are influenced by prediction errors and send predictions to the auditory context.

And the auditory system that further inform the prediction errors.

So this this sort of low level processing of the information that's coming in as we as we as we listen to anything but I think what's nice is that the predictive sort of processing framework acknowledges that we're not, you know, always providing that same amount of,

well,

attention to the auditory system.

attention to every incoming stimulus and in this framework is an explanation of where and how attention.

happens what wine why it's deployed.

And the idea is that.

any agent will look at how reliable data is that how reliable the data that's coming in is in order to see whether they should use that data to in the inferential processes.

This is, you know, known as precision waiting is meant to be carried out by these neuromodulatory cells that are different from the cells that are carrying out the actual transmission of predictions and prediction errors.

But simply put attention is seen as this sort of deployment of high precision or confidence in the beliefs about how observations are related to our actual states, and it's set to enable us.

what we hear to more greatly influence our predictions and the interpretation of the heard sounds.

People have begun to look at this idea of precision waiting and the extent of a prediction error and the extent to which it can be seen in neural data when people are listening, when people are varying levels of expertise and therefore different generative models of the music they're hearing.

people have begun to look at the same time when you know when they're listening, when you know when they're listening, what's happening when they're listening to the pieces of music that are very predictable, very predictable versus less predictable, and they begin begin begun to be able to show that indeed the connectivity within the auditory cortices within STG are reflecting this idea of precision waiting where you're able to.

and then the ability to basically basically basically upweigh to provide sort of more gain on the neurons that are processing incoming information or not.

Of course, the moments were attending but the moments where we're not attending and here as well.

The sort of literature coming out is suggesting that there are ways to account for this so the role of opaqueness has been suggested to be quite important in the meta awareness of our attentional states where a status has to be opaque.

when its underlying processes can be attended to or is visible to conscious awareness.

As such, if you talk about opaqueness and attention and meta awareness, so attentional states are the second order states that allow first order states to become opaque, whereas meta awareness is the sort of third order state above attention allows us to allow attentional states to become opaque.

But generally the idea is that meta awareness is possible thanks to sort of more deep generative models where you have this higher level that's modulating the opaqueness of our attentional states and allowing us to distinguish when we're attending to the sensory stimulus or compared to when we're not.

And if we realize that we're, you know, not attending, we might want to take mental action.

And this has been described as possible or inevitable thanks to this imperative, active inference imperative to choose actions that will bring us close closer to our preferred state.

So if the task we're carrying out or what we're meant to be doing at that moment is listening to the music or we've decided that's what we should be doing, then the idea is our generative models preferred state is the state of attending.

And so if we notice that we are attending while we're attending, this shouldn't cause any surprise, but if we notice that we're not attending, we notice we're mind wandering, this should result in a surprise that under the free energy principle should mean that we need to minimize the surprise and which means that we therefore attend.

So this basically we're seeing how being able to have mental action can lead us to continue this process of attending.

But there's some situations where we want to actually take action, however small to, that we will take action, however small to really engage with, you know,

stimulate in the case of, in the visual domain and maybe eye movements in the case of hearing, it's not always easy to see, but it could be, you know,

talking our ear or, you know, head movements to ensure that we're getting the signal as best possible.

And basically here as well, we see an account of, you know, how so called curiosity, what I'm calling curiosity related behaviors may come about.

So the idea is that these curiosity related behaviors are able to resolve uncertainty through the choice of policies and through us choosing policies that can reduce uncertainty.

So if we, main aim is to reduce uncertainty, then sometimes we need to choose actions that will get us there.

And in this paper, they carried out some simulations where they showed the, you know, a given agent will choose, sometimes choose policies that by exposing them to novel combinations of states and outcomes.

Could on the longer run allow discovery of how outcomes are generated and improve models in the long run.

Of course, curiosity is usually described as a feeling, you know, and it's been suggested that curiosity could be seen as this, what it feels like when we recognize that there is uncertainty out there that we can resolve by taking action.

Just to point out that in this study, in this paper rather, they suggested that evaluation policies may be taking place in areas like the quality computing, which I think is quite interesting given how these have been involved or implicated in the anticipation of sort of pleasure that people have also linked to some extent to that wanting or, or an event or, or a given experience.

And last but not least, I think no account of music listening is, is, is complete without sort of speaking to, you know, its capacity to, to provide pleasure.

And the work coming out of the predictive coding framework or accident forensics suggests that affect isn't just, isn't related just to the magnitude of reward, but to learning dynamics where it's the result of Asians tracking fluctuations in, in how certain their models are.

And it's been put forward to that, you know, if curiosity is the expected rates of reducing or minimizing prediction error than affect could be seen as the actual rates at which we're able to minimize error where this could be faster than expected and lead to this sort of more positive feelings that we can experience in different contexts.

So suffice it to say that predictive coding framework, you know, for very long, we've used it to sort of account for prediction errors, which was a real priority for us in music listening because they're very easy to measure.

And then we've used it to measure, or at least we see these responses that can, we can pick up very easily.

And yes, we have these overarching theories about musical expectancy and the role that plays in our perception and ability to derive emotion and meaning from music.

So predictive coding framework has been very influential as neuroscientists of music thought about sort of music and perception.

But I think what's citing is how it's beginning to give accounts of what's happening when we're attending or, you know, this, you know, distracted or mind wandering how meta awareness can emerge or could take place within this framework, allowing mental action allows us to re-attend if we realized we weren't attending and so on and so forth.

How we might, why, how and why, not how and why, but how, again, curiosity related behaviors could look in this framework and why positive effect might emerge.

And I'd suggest that all of these in some ways are as relevant when we want to answer the question, what is behind this human drive to engage with music?

So we could talk about pleasure, which is what we've tended to do in the sort of neuroscience of music world.

But I would argue that there's all these other reasons which are very much shared with information processing and other domains.

And just as an aside, if anyone's wondering curious about music, how is that possible?

Just to say that we'd just as a proof of concept, we did a few studies to show to see whether people's self reports of curiosity made any sense in relation to what was going on in the music.

And sure enough, whenever they perceived, often when they perceived change, people would also report being a bit more curious about what they were hearing.

We were able to show that people's curiosity ratings when we probe them at different points in a unfolding piece of music was explainable by information content, how surprising that event was, and that this further was interacting with entropy.

Again, to explain the sort of feelings of curiosity suggesting that there was also this precision-writing mechanism going on there.

And just because those two studies were looking only at self-reported music, linking that, or curiosity, linking that to the music.

But because it'd be nice to have another behaviour that isn't sort of self-report, we looked at whether we would see what we thought would be a negative relationship between mind-wandering and curiosity in the context of music listening.

And sure enough, we showed that when people reported high levels of curiosity about what they were hearing, they were less likely to report mind-wandering at those moments.

I could describe the paradigm if anyone's interested.

But yes, indeed, we were able to show that you're more likely to be mind-wandering when you report being less curious about the music that you're hearing.

You're more likely to be mind-wandering if you're less expert, or in other words, have less musical training.

You're more likely to be mind-wandering if music at that moment is quite complex or more entropic.

But we saw that this association between mind-wandering and this tendency to mind-wonder more at high entropy was reduced with expertise.

So the more you had musical training or the more you seemed able to deal with the complexity and not mind-wonder when you heard such complexity in music.

And now just to finish with a short presentation of this model or theoretical framework that we put forward with regard to how engagement with music might look over time.

And again, this was just because after doing some, not the only thing I do, but when I do music neuroscience studies, there's a tendency to use quite short and quite, I don't know, reduced stimuli.

Pieces of stimuli mostly because we have very clear hypotheses and ideally the stimuli are very easy to, you know, they don't have too many confounding things going on.

That was tended to be the case. It's getting better now. We listen to sort of, we use stimuli that are more naturalistic, that go on for minutes.

But in that sort of, in that spirit of what's happening when we listen to a piece of music or several pieces of music back to back over several minutes, in that spirit of what's happening, we decided to sort of put forward this model.

But just to say, you know, what has been said about music engagement and the literature, fairly new term, hadn't been really, hasn't really been used much.

But when people have talked about this idea of being engaged with a piece of music or engaged in the context of music listening, they have tended to emphasize this idea of being compelled or drawn in or interested in what will happen next.

They've emphasized focused attention, so feelings of being actively immersed in the experience to the exclusion of extra musical stimuli.

And they've also pointed out that it has, you know, seems to have temporal dynamics.

It's, you know, something that might change throughout the listening experience.

When we were looking a couple of years ago or a year ago, there wasn't, however, any sort of theoretical accounts of the factors that are contributing to, you know, this continuous experience over time.

And so when invited, we sort of to the special, you know, contribute to this special issue.

We wondered, well, can we build a useful framework of how things like curiosity and attention and mind wandering may emerge and, you know, during the context of music listening in order to see if we could drive research in this area forward.

And just to sort of give a few fundamental claims that we make in this paper.

Well, we suggest, for example, for a start that our engagement with music is going, is cyclical.

It's going through, you know, peaks and troughs.

We are not always engaged.

It's perfectly, makes perfect sense.

We find it very difficult to not mind wander in any, in any time, in any context.

So it's not surprising that this is the case of music.

But certainly we suggest that if there are moments of people suddenly entering into heightened attention or engagement with music, it might be because music's just started.

So there's a capacity to see if there's anything to be learned or interesting way to, yeah.

But it could also be moments within the music that are moments of high surprise and what was up to then a quite predictable context or even low surprise and sort of more unpredictable context, because that can be just as interesting for an agent.

We suggest that, as I said, if there's peaks, there'll be troughs, the moments of engagement, these moments of engagement will be interspersed with mind wandering, moments of mind wandering, which could be explained by the redundancies in the music that preclude any sort of curiosity related behavior or attentional gain.

And then we, you know, I think one of the most crucial claims is that, you know, when we refocus on music after mind wandering, it could be because of what's happening in the music.

Something that may have, you know, that's something that may have, you know, led to attentional gain or sort of curiosity, thanks to, yeah, the novelty or the fact that we, there's something we don't quite understand.

But it could also be because, you know, our capacity for meta awareness has allowed us to see that we're no longer attending and has allowed us to, you know, carry out this mental action where we re-attend.

And last but not least, just to say that, again, a claim of ours is that, as is the case for most music listening experiences, the level of engagement will be, and changes in these levels of engagement over time will be modulated by complexity of what's being heard.

So overall complexity will determine how engaged we are in the first place, but also within that, if we are in a very predictable piece, section of the music and suddenly something more complex happens, you know, that's likely to induce, you know, curiosity or sort of more attention,

compared to, conversely, if something more simple and unsurprising, maybe what actually triggers our interest and engagement.

So basically saying that, like I alluded to earlier, it's not as simple as complex or sort of surprising will always trigger interest and engagement.

It's quite, it can be quite nuanced. But yeah, also saying, we also, you know, argued that expertise will of course play a role, because everyone will have their own generative model.

In the case of someone who has lots of musical expertise, this might be a very refined and sophisticated generative model that allows them to make better predictions and tolerate the uncertainty or entropy that might exist in a piece of music at that moment.

And last but not least, we'd argue that the listening situation context really matters, tells us whether or not we have the capacity to deploy attention to the music, instead of say, I don't know, announcements being made about the train that we are hoping to get on in the next five minutes or so.

So just to summarize what I've said today, the question was, what is behind this drive we seem to have as humans to engage with music and I gave this sort of an account that has come out of the work that's being done in sort of the neurosciences of music, where, you know, we started looking at music perception,

we looked at pleasure and we're beginning to, or musical reward and we're beginning to tease or put together a story about how pleasure is can be explained by these frontal temporal connections that are interfacing with the reward network.

I think what's quite nice is that we're sort of seeing the work that's been done in animal models showing that information can be a reward and beginning to try to conceptualize music a little bit in these terms, you know, where perhaps a sensory prediction error is beginning to become something that can actually result in a reward prediction error, which would of course make it easier to interface

to be able to use the conceptual systems in a way that we're not always immediately obvious with music, a very abstract stimulus.

I've also provided sort of, or reviewed some of the predictive coding literature or active inference literature that I found really compelling as I try to understand what might drive changes in our engagement with music over time, try to see how, you know, using this one principle, you could explain lots of different behaviors,

lots of different behaviors, you know, are probably going on as we engage with a piece of music, and just not so much suggesting that pleasure is a consequence, but, you know, suggesting that it's part of several different processes.

It's not the only reason we choose to engage, even though we might learn that music is pleasurable, but we engage with all these other reasons first.

And just to say that, you know, these are accounts that are very much increasingly converging lots of the work we do in the neurosciences.

And when, for example, I'm doing it, it's not, again, the work I do, but when I'm doing it, I definitely have an eye on what's coming out of, you know, this literature, and likewise, I think, you know, several people active in this are working together with neuroscientists.

Neuroscientists, I think, there's clearly, you know, increasingly converging, I think there's still a lot of work that needs to be done to link some of the optimization processes, which, you know, we hear talked about in that literature with the neural substrates and processes that we are able to look at with

Neuroscience and music studies, but the hope is that if we start putting forward some frameworks that are more closely informed by this literature, perhaps the more we can drive empirical work that can get us closer to these, you know, accounts really, not just converging, but being, you know, really overlapping and helping us understand this question, which is, you know,

what is behind the human drive to engage with music.

So I'll stop talking. I don't know how long it's been, but I'm looking forward to thoughts and discussion.

Awesome. Thank you. What is behind the human drive to engage with music?

So I think a desire to learn about the world that we're in, or to hear desire, just a need to improve our model of the world. I think this is rewarded. And music, we very quickly learned, is a way to learn and be rewarded at the same time that became something that

we do, you know, we do, you know, as much as, you know, an average, you know, a fifth of our waking hours.

Yeah, it's quite a lofty question, but I would argue it's, at some point we learn that music is rewarding and that's why we turn it on. But I'll say, ultimately, so is food and I don't know, there's lots of other rewards, something special about music.

And I think it's this unfolding of abstract information that somehow, you know, can capture our attention, perhaps distract us from other concerns that are more basic and yeah, still somehow reward us for that distraction.

Wow, very interesting. Like it has both the element of pragmatic value, especially like songs from our childhood or a period that we're familiar with listening to the same exact recording, or to a cover song or to some other version or like a live version.

There's sort of a familiarity, which can be like a pragmatic value.

And then also you're, you're really getting into the epistemic value and the fact it is learning.

And then there's so much variation, like what genre, what song? Do you listen to new music? Is it an playlist assembled by an algorithm? Is it something that somebody recommended to you? There's like so many layers of learning, even if it's a simple sound.

Yeah, I think so. I think that's what's interesting about music. You can sort of study what's going on as people are listening, but you can also study their actions, their behaviours, what they go to see, what they choose to listen to, what they buy, what they're willing to spend their money on.

And we, with curiosity is just an idea. We explored it also with regard to what people, we sort of look not just at curiosity as a state, but as a trait and shown some, shown that actually, yes, generally being someone who's trait curious, who will seek out new experiences, explains, you know, how much you'll report enjoying the experience.

How much you've never heard before from a culture you've never heard before and so on and so forth. So I guess that was a bit of an aside. I guess there's a lot of individual differences.

I guess there's, you know, other more general traits, you know, other more general traits.

And some people talk about personality traits like openness will also predict how much we will show these kinds of more exploratory behaviour, or how much we'll just exploit the music that we know really well.

I was really struck by the definition of curiosity that you used something like uncertainty we feel we can resolve with action, not just to know that we don't know, is sort of a condition of curiosity, or it's a relative of curiosity, but but to know we don't know, or to feel we don't know, and that we could do something active to epistemically forage.

And then music is a special situation where the action that you can take, it's like, what will happen in this techno beat drop? Will it be two measures or will it be four? But the action to take is to wait.

So it doesn't require another over body action or a mental attentional action other than just to attend and to let it unfold.

So it plays out in time with curiosity with sort of like a passive action choice, choice not to act not to turn it off.

In a way that other information reducing situations that are they're modeled as like going out of your way to do something to resolve uncertainty.

Yeah, that's a great point. I think there's lots of paradigms.

Well, the best part, some of the best paradigms for looking at curiosity is, you know, how much you're willing to spend time pay, you know, attend to pay attention, to actually find out what's going to happen to acquire that knowledge and exactly right.

I think one could build and I just haven't got around to it paradigms where you're looking at just how much people are willing to keep listening instead of switch over, you know, almost as though it was the radio.

Yeah, people also looked at exerting effort, but those tend to be more artificial.

Yeah, they've used these kinds of paradigms to see just how much you're willing to do to get that information.

But I agree. Yeah, we don't have many overt actions that are observable with listening to music other than just not turning it off.

Yeah.

Yeah. And that points towards this extended engagement, like certainly from little snippets of sound and music, certain aspects of listening and attention can be explored.

But then there'd be other dynamics that would only start to come into play, like over multiple minutes, like multiple songs.

Mm hmm. Mm hmm.

Mm hmm. Mm hmm.

Like I'm thinking about being at a concert.

We're curious what song they're going to play late in the set or if they'll do an encore.

Also, though, we prefer to go to bed early.

So there's more traffic. It's getting late.

So then it's like that's a revealed stay or go of how much do we want to resolve the uncertainty as another cost starts to another trade off starts to build up.

So that's happening at a slower level than like rhythm detection over a three second sliding window.

Yeah, definitely. And I think that's where it gets really interesting and quite complex.

I like the literature coming out about how curiosity is contagious.

And, you know, that, you know, this idea of the social aspects of information foraging or, you know, caring about information.

So with regards to, you know, why you stick around and don't leave, maybe it's so you have the complete picture of what that concert actually turned out and you want to be able to share that or you certainly don't want to miss it.

Other people are sticking around. So, you know, you're thinking, well, maybe they're right.

This is something to stick around for.

And yeah, I think all of those different levels probably interacting, I guess, if you've got lots of boring tracks towards the end, you might be more inclined to throw in the towel and leave before it gets really bad traffic to be really bad traffic.

But yeah, ultimately, it'll also depend on your individual differences.

And at that point, I'm just sort of need for cognition or, you know, knowing everything, but also, yeah, other things, how much you want to get your money's worth.

Yeah. Another interesting piece was the two streams. So that had a lot of similarities in the presentation with the visual what and where.

Yeah. And it was almost like there was a what and a when. So maybe could you restate that or how does that when this come into play or how is that separated out from an essentially temporal process?

Yeah. So, yeah, very simply. Yeah, it's very much borrowing from what had been put forward in the visual domain and lots of music research, music neuroscience research has borrowed from what's gone on in vision and the vision sciences of vision neurosciences.

Yeah. And yeah, the sort of what pathway is this ventral pathway in the case of the auditory system. It's the areas that house the auditory cortices feeding into the inferior temporal, inferior frontal areas where, which are generally involved in sequence sequencing and storing sequences and using those to make predictions as to what the next event in the sequence might be in any given in any given sequence. So that's the what predicting what's going to happen and, you know, setting predictions down the hierarchy about what. Yeah, what that might be with the when music isn't just, you know, what's happening. It's when it's happening. And so the when paradigm, this dorsal. Sorry, the when pathway. This is the dorsal stream. That's unseen.

Information travel between these again core auditory areas and the superior temporal gyrus and motor and parietal areas that are involved have been shown to be involved in sort of temporal processing. So lots I didn't show much work on rhythm for me to processing because I tend to work in pitch and that's why I meant by sometimes I'll say we can be very reduced. It's because we just want to say look at probabilities in the pitch domain before we, you know, complicate it by adding

lots of temporal information. But yes, if someone were to review that literature with you, you'd see that motor areas are very much involved in our processing of time in music.

in our processing of time in music. And so this dorsal pathway is basically doing everything from helping us predict when something might happen in a piece of music to helping us.

Yeah, well, yeah, predict when something might happen, but even helping us make transformations or auditory motor.

or experience sort of the motoric actions that could be seem to be attached to that listening experience. So if you're a musician knowing exactly.

It's underlies the sorts of bodily experiences of the movements you might make, even though you're not making those movements. So the action, you know, observation, literature, and their neurons as well.

So these two streams are clearly very important for just processing sound as musical sounds as they evolve. But interestingly, in this book I mentioned, where Robesitore is trying to put lots of things together, he makes this interesting observation, which is that the dorsal stream might be talking more to the dorsal striatum, which is involved in the anticipation of the

chills, which again is this kind of like you're waiting for something when am I going to get this extremely rewarding moments that is the modulation or the drop.

And that, you know, prediction about what that drop is going to be is meant to be observed by the ventral system, which he, you know, they shown in a number of lots of different studies is almost related more to this ventral striatum.

So yeah, it's, I think it needs to be a little bit more carefully.

Yeah, studied, but it's just another way in which there have been people are making attempts to try to understand how the perceptual system might be interfacing with the reward systems in a way that we've, you know, we can see anatomically presented in our brains.

How are you seeing AI analysis and generation of music changing our understandings and habits and behaviors and listenings?

I don't know.

I don't know much.

There is, I spot more and more, not surprisingly, papers or talks on this, around this topic.

I mean, AI is impacting so many aspects of our lives.

I think even before AI, there was streaming, there was lots and lots of music that we could expose ourselves to.

It could just play the whole day compared to, you know, when we would have to buy CDs and I don't know, turn them on and so on.

So I think there's some interesting behaviors perhaps that have already begun to emerge as we have more and more access to music.

I think, you know, what's interesting about AI is that, you know, it might be in some cases, very little human, very few humans involved.

And I think, you know, there's some papers out there already published, I want to say 2007, but it could be 14, at least 10, if not 15 years ago, that have shown that if you thought you were listening to a piece of music that had been composed by an algorithm, you showed a lot less activity in the theory of mind network as if you thought this piece of music had been composed by a composer.

So the stimuli there were sort of atonal sort of serial, serialist compositions, which seemed plausible, you know, composed by an algorithm and also by a composer.

Now, of course, AI is very convincing and it'll be interesting.

I think it'll be harder maybe to detect or tell whether or not a human was involved, very hard indeed.

But I think my suggestion is that once we recognize that there wasn't really a human involved, we might show a drop in engagement with and that in that sort of social way that music can be engaging where we try to understand what the artist is trying to tell us where we want to feel with the performer and the emotions they're expressing.

And without that recognition, I, you know, I think we're just listening to music and we'll engage, you know, to the extent that we find it interesting or stimulating or pleasurable, regardless of, yeah, where it came from.

Yeah, that's interesting.

Yeah, that's interesting.

It's almost like the ability to generate images, sounds, et cetera, is helping to articulate out what are the aspects of this image that have a primary saliency?

Like whether for pragmatic value aligning with what we like to see or hear or the epistemic, the learning, and then separating or factoring that off from the social and the relational.

Like this drawing was made by this person.

So I like to look at it.

It has a saliency and also those pragmatic and epistemic elements, not only from its actual composition, but because there's like more knowledge about the generative process that led to its construction.

So then those paths may diverge more and more as like new genres of very salient music.

Like I thought about on your plot with the affect, are there genres or subcultures in their listening of genres that it's just like it stacks all the positive affects?

And then that made me think a little bit about the sped up mashup remixes that change songs.

And it's like songs and nostalgia at a very, very fast rate.

Interesting.

And so it's like, it's just like a few notes to invoke something and then onto the next one.

And it just this kind of moving on through novelty and nostalgia and what you do and don't know.

And it's kind of like, it's so compressed and trying to have every part be really hyped up.

And then other pieces that maybe like considered from the album composition might have like a slow song.

And then it's like, I think that's a long song that's not trying to be so salient even, or it's giving a contrast that sets something else up more deeply.

Yeah, yeah, no, I, yeah, it's great.

I love that.

I love the sort of mashups.

I was like, yeah, feed me all this nostalgia.

I think that's so there's our interest in feeling with the other, Ein Fühlung, empathy, that's meant to, you know, it's suggested that actually it's a fact that the word for empathy came out of how we described our experiences of the arts with feeling into the arts.

So there's this social aspect to listening, people talk about surrogate, or just this idea, even if we don't, there isn't anyone there.

And it's just a recording, we're listening with our headphones on the tube.

We still imagine this person, this, that, but I think the nostalgia is really interesting because we're almost like empathizing with our previous selves and our feelings and emotions there, which have been so, which music is so able to bring back to us, you know, what our

priorities and our values were in those, you know, those stages when we were listening or just, you know, what fun times or who we were hanging out with at those moments.

So I think, yeah, that's, that's really interesting.

I mean, and like you said, is really trying to do, is trying to really tap into this desire we have to use music to connect with ourselves.

Yeah. And then yes, something that's really slow going, you know, very minimalist really does set you up then to find the most tiny deviation, extremely salient and interesting and worthy of pondering or just exploring that experience.

And they're very different things. And I think we'll, we'll seek the, these kinds of music out at different, you know, moments in our day, in our lives.

Some of the music will just constant, will completely, will capture and hold your attention.

You know, we've seen with this idea of the attention economy where, yeah, we only have so much attention to, to share amongst the things that might compete for our attention.

And people use that to look at how, you know, how that might have been influencing, how this principle might be influencing how music is composed and has been composed popular music in the last, you know, 50 or so years.

And showing, you know, and showing your music is louder, you know, the hook comes in quicker, and, you know, and so on and so forth.

And so that's, you know, an example of this tendency for some composers to try to hold, to capture and hold our attention because it's so easy to just click next on Spotify.

Where, yeah, others conversely, probably are trying to react against that or their aesthetic values are that music should never be that engaging and there should be time to ruminate or just to follow the form, you know, for as long as, you know, you'd like without having to process the next thing.

And then the next thing and then the next thing.

Yeah, different affordances, I guess, the music has for any, any listener.

Yeah, that's very like media is the message for a record to begin with 10 or 30 seconds of silence or the slow buildup, when there's already like the few extra seconds of just the record tracking.

It's not like you're going to switch the song.

But then if there is a way, if it's a playlist, having 10 or 30 seconds in the song at the beginning or the end, it might not be played like in a public space or on a run where you might be expecting continuous music and then that changes the technical culture of listening.

And again, that's related to this extended engagement, building on the pitch and the rhythm identification.

There's still these questions about longer term listening.

It's kind of like the dark room.

How well, okay, if we're bounding surprise, why not the dark room, etc.

Not to go into this, but it's like, but what about the movie theater room?

Or what about the sound listening room?

It's a not, it's like a little bit of an, it's a enlivened room, it has stimuli.

And then those are the situations that the extended engagement is looking into.

But how long do you stay playing a video game or listening to music or playing music when other things could happen?

And then it's not just an inner analysis of music.

It does start to invoke like, well, you can't just do this for 24 hours.

So there has to be some stopping time where even the most passionate person is still going to like take a break or something like that.

Hmm.

Yeah.

Yeah.

And I think what's interesting is, you know, most times we hope we have that ability to decide what's happening.

And maybe we can prepare ourselves or tell ourselves we can do this tomorrow as well, or, or savor the experience.

So, you know, whenever I've watched a moving movie in the theater, most times I'll just sit and watch the credits and listen and just kind of reflect.

It's very annoying when you watch a movie on some of these Netflix and before they've even done a few seconds of the closing credits, they advertising the next.

I think we probably have learned to, you know, this savoring of what we've just experienced and, you know, letting it tail off in a way that's, yeah, we can manage as opposed to it suddenly being rudely interrupted.

And I think that's probably why we can, you know, even the most passionate listeners able to turn it off and be like, okay, now I actually have to get some work done or now I have to, you know, go to bed.

Um, because we can manage that and tailing off most times if we're lucky.

Yeah.

And the space within and at the end as a kind of enculturation, like in classical music, there'll be a pause.

Nobody will clap or you'll hear one person clap.

It's like, oh no.

But then at the end and then clapping, it's a motor, a joint motor activity that signals, okay, it's really the end.

And so then that there's like, you, you mentioned, um, the expertise increased the focus and the epistemic component.

Possibly if for nothing else, like I want to see how this book ends.

I want to see how, I want to see how the structural elements and what I've heard so far relate to how it ends.

So it's like all because of knowledge of the structural form or the genre of the murder mystery or the, this kind of song, that kind of song, there can be a structural curiosity that keeps one engaged.

Even separate from any other kind of curiosity, just like as a craft, how will the author do this?

How did the composer bring this all to a close?

And then that gives this epistemic, um, arc that, that is informed by the genre.

Yeah.

I'm reminded that, um, the study where, uh, study, I keep saying study, these empirical work, or the, the paper where, um, just some colleagues talk about sort of, sort of,

some inference and curiosity, and they actually talk about insights as well and talk about sort of how, even without extra information, sort of just restructuring what you, what you've acquired, what you've learned, can lead to these moments of insights.

And I guess when I'm sitting there, taking in everything I've, you know, just read, you know, nothing worse than reading a book and then having to, that's it.

And, you know, go off and do something, you know, not having that moment to sort of think some more.

And I think it's because there's probably insights to be gained only after you've come to that final end, which I'm sure was carefully crafted by whoever, um, you know, created the work and yeah, just time to process further and, you know, gain some further insights.

Um, there is that, yeah, what's happening during the artwork for this time that's needed to, to establish what was actually learned, you know, once everything had come together.

And it makes me think about the John Cage type concepts of music being time, time bound, whereas a painting or visual art, it doesn't have a constraint so much or so similar on time.

So, at the time, the museum observer could be there for a split second or for a long time, but then there's a time boundness.

So it's sort of like, well, there's, um, a canvas of just one color.

And then that is like the 433 song that's a duration.

Oh, there's nothing there.

Oh, well, there's the ambient sound and there's the structural elements voided of traditional contents, just like the canvas kind of cleared of its traditional contents.

And then how, how those kind of works like explore limits of art.

However, I don't have blank canvases around me.

Most people don't listen to 433, um, when they're going for a run.

Yeah.

So it's, there's still is, is preference distributions overlaid.

It's not just about having like this, this, um, genre defining like meta art piece.

There's also just like the normal day to day.

Um, yeah.

Yeah.

I mean, I think that's where people talk about the context.

So, um, yeah, you need that work needs to be in a museum or a gallery, um, or in a concert.

Um, and that's, um, that's the case may be, um, people talk about the aesthetic attitude and like distance, or, um, I think there's that element of, yeah.

The extent to which you can employ cognitive resources to this, not extremely urgent problem.

Um, and there are very few situations or circumstances where you can just, sorry, in daily life, where you have that time to, um, you know,

Um, to do that.

And so setting is so important in addition to these individual differences and, you know, how much we want to spend our time doing that, that sort of thing.

Cool.

Well, what are your next moves or what are your directions for research with music or for anything?

Um, I've always, I, I, yeah, I got to sort of work in very neuroscience places and other places that were, you know,

do neuroscience, but we care about these lofty questions around this experience.

Um, so, and I, I, at some point I got interested in, I think, starting with surprise and, you know, just electrophysiological responses to those, how that, you know, could lead to curiosity, this, you know, extended engagement.

But I, I guess I've been interested in other fairly epistemic emotions.

So, um, uh, or is one that we've been exploring recently, um, uh, with students.

Um, and the reason is, um, well, it's one of those emotions that are very hard to pin down as a, an experimenter.

Uh, we're going to try to use, we've been trying to use, um, VR to create more naturalistic settings that people might hear music or, and even then begin to compare what.

Or induced by music might look like compared to or induced by, I don't know, nature or the visual world and so on.

Um, and yeah, trying to understand a little bit more about.

Uh, desire for learning or understanding.

How, you know, experiences like wonder can come out of abstract stimuli like, um, music.

And yeah, I really bridge this link or better understand this link between knowledge seeking and, um, aesthetic transcendental experiences that we can have with the arts.

So there's that kind of work being done.

And yeah, other work, we sort of look at the images, the music can conjure up, but I'm really interested in, you know, predictions or engagement at those levels as well.

So in a way, some of the accounts I've given today are very much like, Oh, and you care about the next note and what's going to happen.

But sometimes you're imagining yourself, um, and others and personal memories, uh, mixed up with all of these, you know, um, thoughts that are more related to the music as it unfolds.

And inevitably, um, it would be interesting to understand a bit more about that level of engagement with music.

So we can see how it's either explainable by some of these principles, or at least, um, needs to be considered, um, alongside.

Um, yeah, I think that's the most relevant work that I'm doing, planning on doing to the work I presented today.

Cool. Well, thank you very much for joining. Super interesting and look forward to talking again.

Thank you so much, Daniel. Thanks. Bye.

Bye.